Biogeophysical and Atmospheric Feedbacks of Cocoa Expansion with Enhanced Weathering in the Tropics

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Background Story

The Vulnerable Hero: The Cocoa Tree

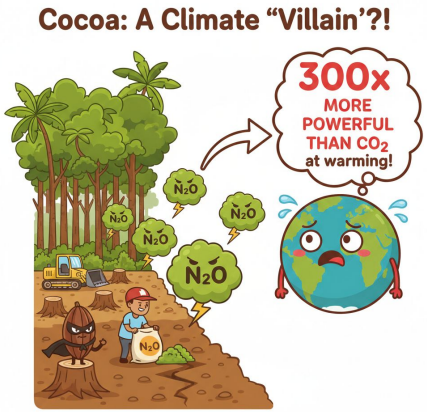
- **only grows in specific tropical regions**

- **terrified of climate change**

Plot Twist

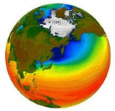
Cocoa can be a bit of a "villain."

- When forests are cleared for cocoa
- soils are heavily fertilised
- the ground can "burp out" nitrous oxide (N_2O) → a greenhouse gas around 300x more powerful than CO_2.



The Problem – A Broken Model

How Do We Predict Climate Change?

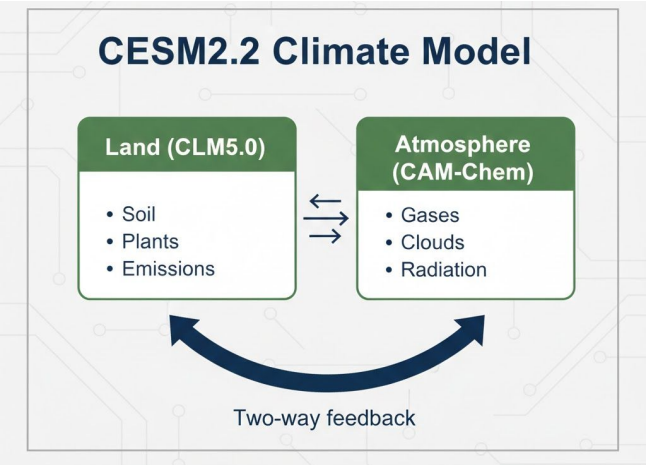


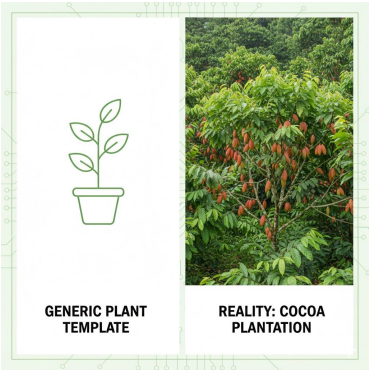
giant computer models: Community Earth System Model 2 (CESM2)

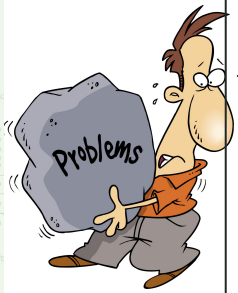
simulate Earth's climate system → 2-way feedback: land & atmosphere

prediction

plants are represented as templates → **Plant Functional Types (PFT)**







The cocoa template in the model is inaccurate

- It's a generic guess, not based on actual cocoa characteristics.
- The model doesn't understand how cocoa actually behaves
- **This means climate predictions involving cocoa are unreliable**

Why Does This Matter? Why Improve the Cocoa Model?

Climate Science

- Models predict Earth's climate future
- Inaccurate plant representations = unreliable predictions
- Better models = better science

Agriculture

- Cocoa feeds millions of smallholder farmers globally
- Climate change threatens cocoa production
- Models help us understand these risks

Earth Systems

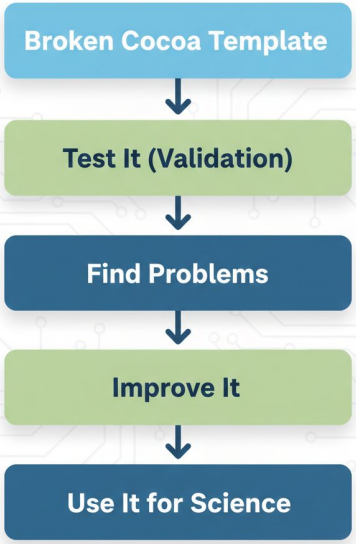
- Plants affect atmosphere (water vapor, gases, clouds)
- Atmosphere affects plants (rain, temperature, sunlight)
- These are two-way feedbacks, must be accurate to model them

Therefore, improving one plant template in one climate model is foundational to all of this.

The Research Question

"Can I build an accurate cocoa template for this climate model, and how do I know it's accurate?"

The Approach

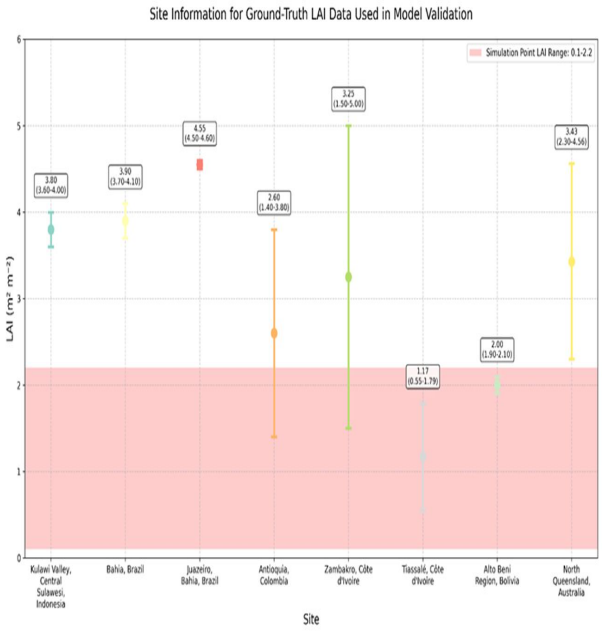


- Test the current cocoa template (run the model)
- Compare it to real-world data (see if predictions match reality)

Find where it goes wrong (identify the problems)

Work to improve it (refine the template)

The Validation – Comparing Model to Reality



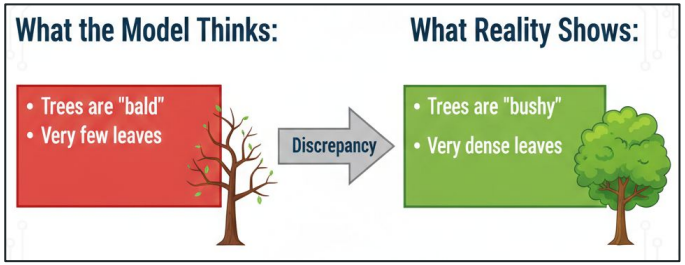
Testing the Cocoa Template - How?

- Compare model predictions to real-world observations (literatures)
- Used published **Leaf Area Index (LAI)** data from **8 cocoa regions globally**
- LAI = measurement of **how leafy a plantation is** → ***"How much leaf coverage does this place have?"***

MODEL PREDICTS: 0.1 - 2.2 m^2m^2

Gap: ~ 50% underestimation

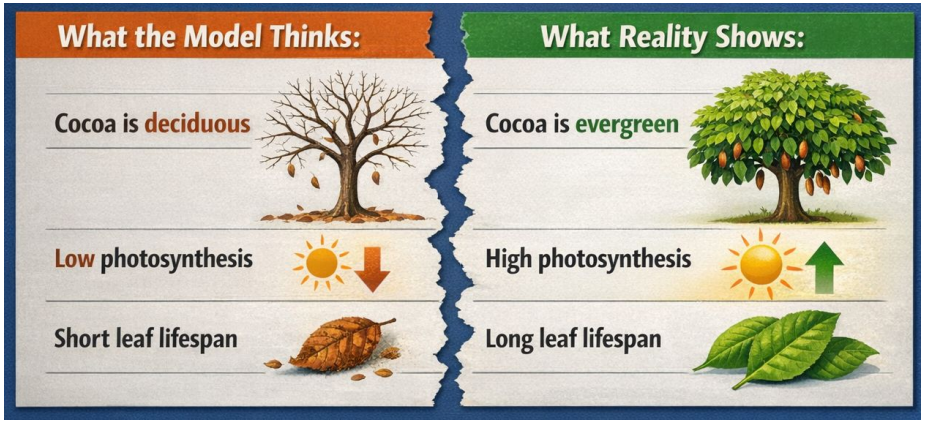
REALITY SHOWS: 3.8 - 4.6 m^2m^2

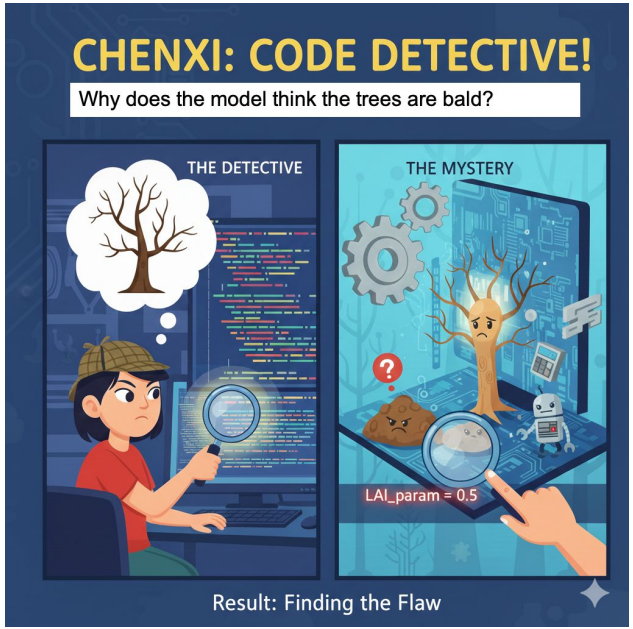


Conclusion: The cocoa template is fundamentally inaccurate

Why the Model Failed – Understanding the Problem

Why did the model get it wrong?





The cocoa template has fundamental flaws:

- **Phenology**: Model marked cocoa as "deciduous" (loses leaves seasonally)
  - **Reality**: Cocoa is evergreen or semi-evergreen (keeps leaves year-round)
- **Photosynthesis**: Wrong rates borrowed from other plants
  - **Reality**: Cocoa has different photosynthetic characteristics
- **Leaf characteristics**: Inaccurate representation of leaf lifespan.
  - **Reality**: Cocoa leaves persist longer than modeled

What Comes Next – The Path Forward

Current Status: Validation phase complete

- Discovered the cocoa template is inaccurate
- Know what the problem is (~50% LAI underestimation)

Next Phase: Refinement

- Analyze literature to understand cocoa characteristics better
- Refine the cocoa template parameters
- Re-test the model against validation data
- Timeline: 2025-2026

Future Goal: Use the improved model

- Once she trusts the cocoa template, apply it to research questions:
  - What happens when cocoa farming expands?
  - How do different practices affect climate feedbacks?

Why This Matters – Key Takeaways

- **For climate science:** More accurate representation of tropical vegetation in global models
- **For agriculture:** Better predictions of how cocoa systems respond to climate change
- **For Earth system understanding:** Improves coupled land-atmosphere models generally
- **Broader relevance:** Framework that could apply to other tropical crops

Building the RIGHT TOOL so that scientists can reliably answer bigger question

Thank you